

Introduction

With the increasing popularity of online taxi hailing services, Uber has become one of the most popular online taxi platforms, gaining recognition from passengers and replacing traditional taxis due to its convenience and flexibility. In Australia, the vast geographic area makes public transport unable to meet all the needs of passengers, Uber has become the popular choice for daily travel in urban area.

This report analyses the key factors influencing the price of Uber rides in Sydney by analysing a comprehensive dataset of Uber trips in Sydney in 2021. By analysing these factors, it can help passengers to understand the pricing mechanism. The dataset was collected from Kaggle, which published by Eugene Koutiloff, and contains 30,195 records with 19 detailed information on trips date, distance, fares, pickup and drop-off locations, etc.

Our analysis focuses on four main aspects:

1. **Demand over time** – identifying trip demand per 2 hour during a day through bar charts.
2. **Geographic distribution** – mapping pickup locations to find out the geographic distribution of orders by using heat map.
3. **Distance vs. fare** – using scatter plots to examine the relationship between travel distance and uber fare.
4. **Duration vs. fare** – analysing the relationship between driving time and fare through scatter plots.

By visualising and interpreting these factors, this report aims to explain the influencing factors behind Uber pricing and contribute to a better understanding of urban travel in Sydney.

1. Data Cleaning

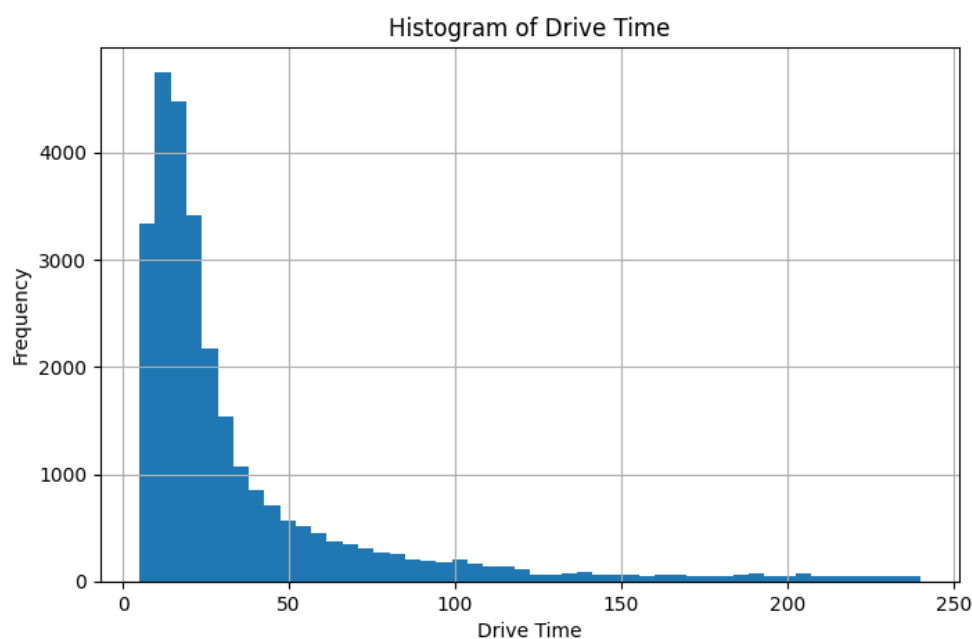
Prior to further analysis, a series of data cleaning steps were performed to ensure the quality and reliability of the dataset. The original dataset contained 30,195 Sydney Uber trip records, including 19 columns, part of them were irrelevant or contained incomplete data.

Firstly, we removed columns such as “Trip ID”, “Pickup State” and “Pickup Country”, which were not necessary for our analysis. These variables either contained redundant information or did not provide meaningful insights for our analysis.

Secondly, we handled the missing values in key columns such as “pickup Suburb” and “Drop-off Suburb” by removing any rows with null values. This is because the suburb data cannot be replaced with mean value and the proportion of missing values is insignificant. In addition, we converted the "Pick Up Datetime" and "Drop Off Datetime" columns to date-time format to support the next step of time-based analysis.

To ensure that the dataset reflects real Uber trips, we filtered out records with invalid or extreme values. Specifically, we:

- Removed trips with a driving time of less than 5 minutes.
- Excluded entries with non-positive values in "Drive Distance Km" or "Uber Fare".



There were 28,294 rows of data remaining after filtering by condition. We re-examined the histogram and found that fewer trips longer than 60 minutes, which explains the large difference between the 75th percentile and the maximum value in the descriptive statistics and may have affected the results. Therefore, we further restricted the dataset to trips within 60 minutes, narrowing the dataset to 23,717 valid observations. The

range of driving times from 5 to 60 minutes is more representative of the daily demand of Uber in urban.

The final cleaned dataset provided a reliable and realistic basis for subsequent visualisations and statistical analyses, ensuring consistent and high-quality data was available to all team members.

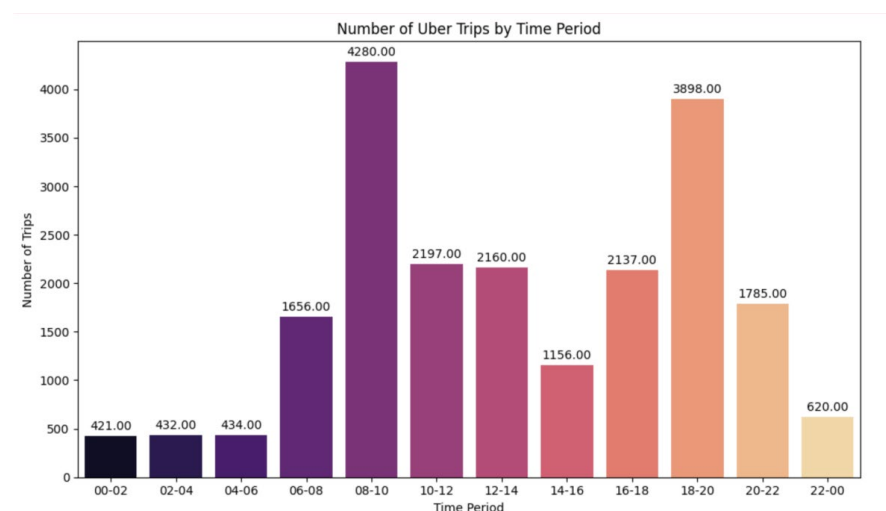
2. How Times of Day Affects Uber Pricing in Sydney

This section analyzes the patterns in Uber fares that exist between the hours of the day in Sydney. The objective of this analysis is to understand how fares vary with respect to the temporal influence of demand, traffic conditions, and socially influenced patterns like school zones and commuting hours.

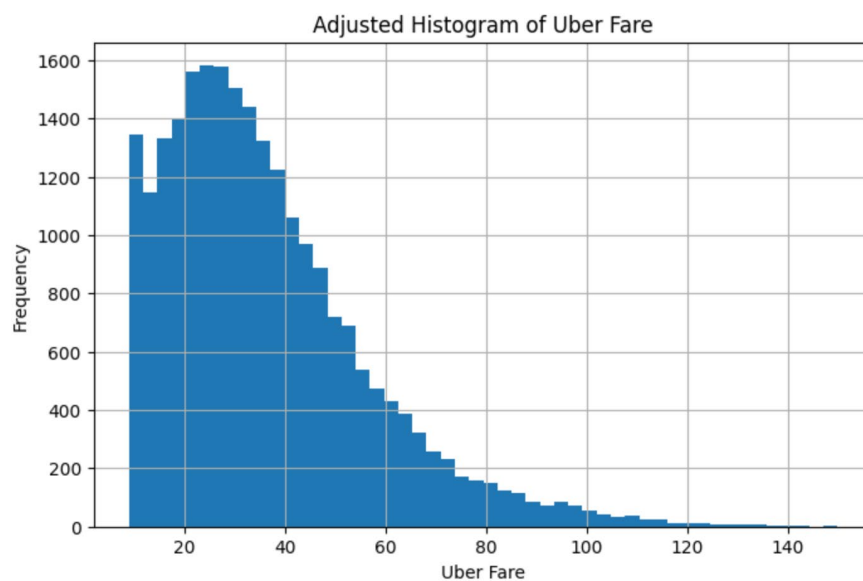
2.1 The relationship between time and order quantity

As a preliminary analysis, all trips were first parametrized into blocks of two-hour lengths. In general, more Uber rides occurred in the block between 8–10am and 6–8pm hours. The peak usage hours for Uber in Sydney correlate closely with peak commute hours and indicate higher demand for user travel during these periods than off hours. Additionally, Uber fares do not exhibit the same non-linear shape over the subsequent blocks and hours of the night. After 10pm, Uber use significantly decreases and does not increase until 6A.M. This indicates it is very likely that the lower Uber usage reported corresponds to lower passenger activity and traffic congestion is generally much lower during these hours too.

2.2 Why choose median rather than mean



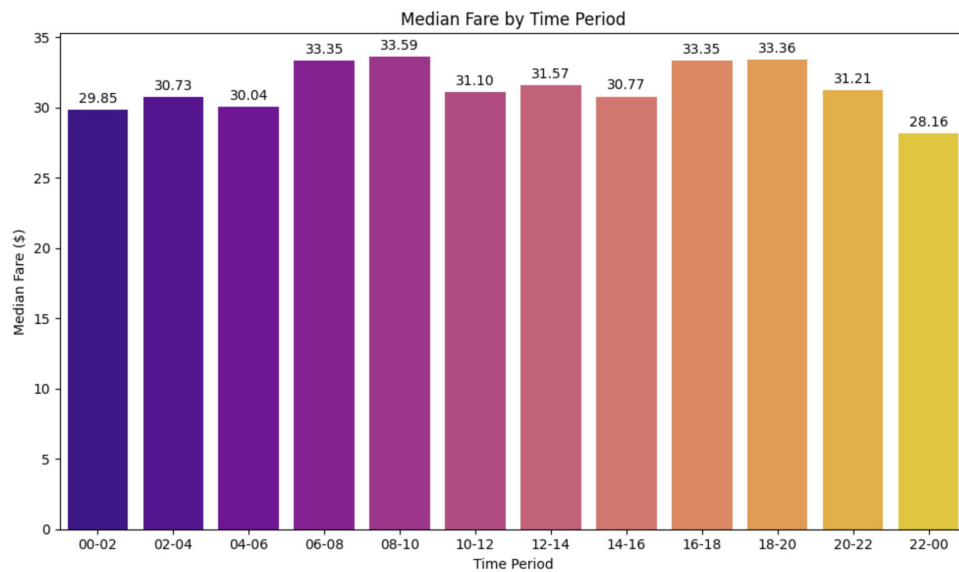
Another major methodological decision was to use the median instead of the mean to present the distribution of fares per time slots. Typically with Uber fare data, data are skewed to the right, as the majority of rides are not expensive, however, there can be a small handful of very expensive trips due to long total distances, and common driver delays related to traffic (the Apartheid hours) and surge price for Ubers. The mean can result in significant issues for data reporting, since outliers impact the mean. Instead of comparing trips through the mean or arithmetic average with respect for hour blocks, median values, which is the 50th percentile of trip data, is an important descriptor of what most users will likely pay for their fares.



2.3 The relationship between time and median price

When assessing the correlation of fare levels, we discovered that median fares were greater in the time bands 6–10 AM and 4–8 PM, which encompasses the school zone periods (7–9:30 AM) and traditional work commute periods. These bands also are the most common periods for Uber to be used, the fares are greater due to an increase in demand, congestion of vehicles on the road, and possibly lower availability of drivers. In addition, after-school times, particularly the 2:30–4:00 PM band, likely contributes to additional demand on the system. In contrast, time bands between 10 PM and 2 AM were less expensive and more stable. The benefit of balanced driver-passenger availability likely contributes to this experience. That is, both supply and demand on the platform are low relative to each other and are more stable and consistent. At night,

fewer vehicles on the road also can lead to a decrease in average trip time and fares as well.

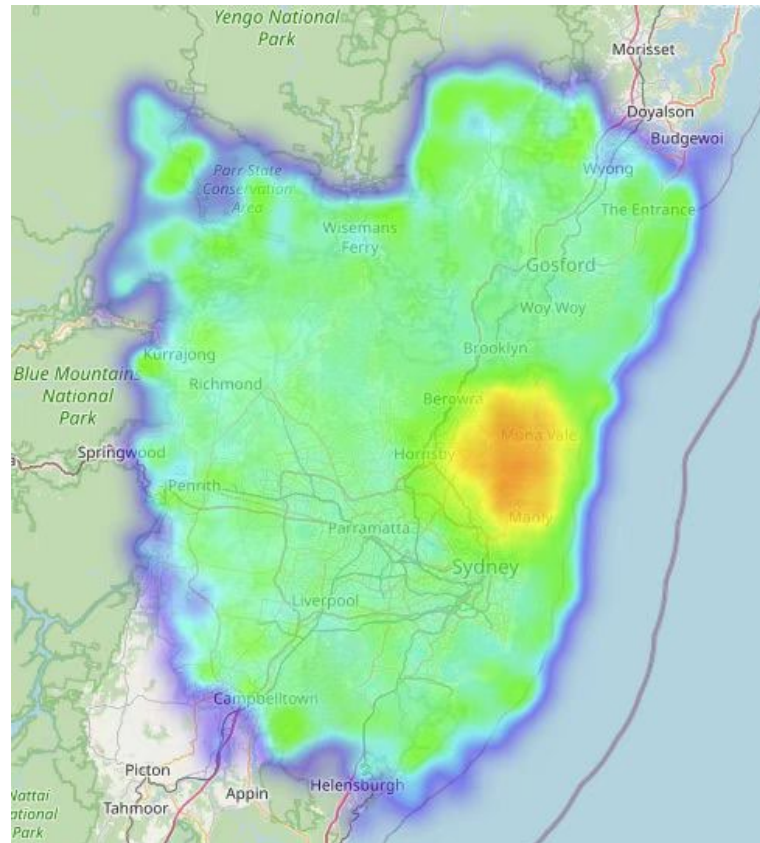


2.4 Conclusion

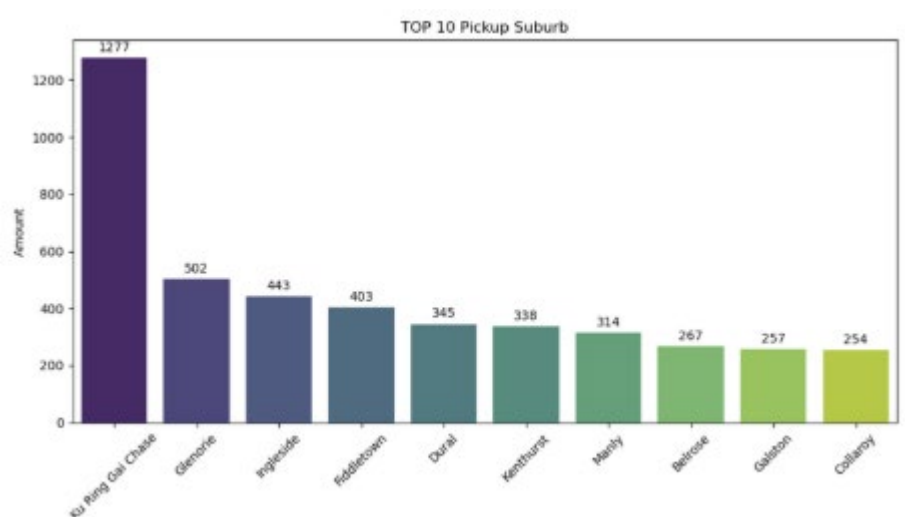
Thus, overall, our study affirms, in the context of Uber pricing in Sydney, fares are directly related to the time of day, in turn, driven from patterns in human activity and sufficient congestion on the roads. When demand is high, especially in time bands that coincide with school zone periods or during traditional work commute periods, fare rates rise higher. Conversely, time bands when demand is low, especially later in the evening, would indicate lower rates. This new understanding can help passengers decide when to schedule future trips in order to minimise costs, and can give Uber insights to add to their future algorithmic based, dynamic pricing decisions.

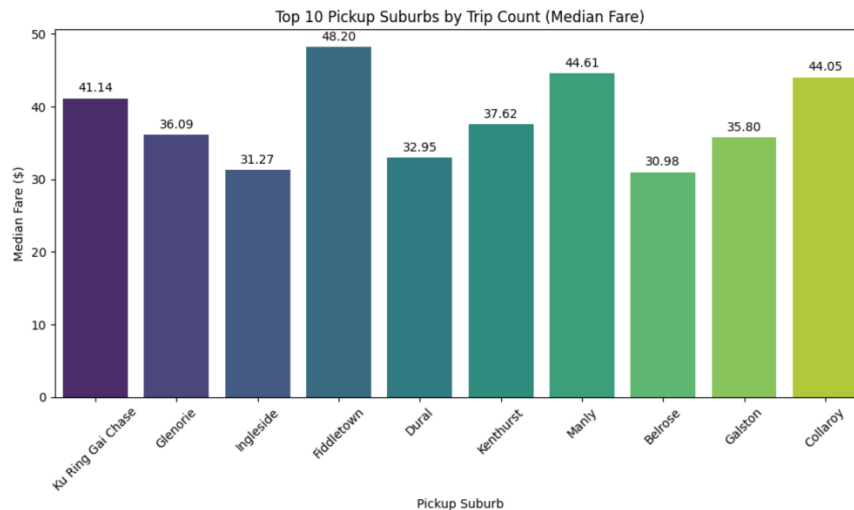
3. How Geographic Location Affects Uber Fares

This section analyzes how geographic location influences Uber fare prices across different Sydney suburbs.



To investigate this, we first plotted a city-wide pickup heatmap, which clearly shows that the highest ride densities are clustered in the CBD and its immediate surroundings, while distant areas record far fewer trips. To ensure our analysis is representative, we then selected the ten suburbs with the highest trip volumes for further study.





3.1 Remote Suburbs with Higher Fares: Ku Ring Gai Chase & Fiddletown

From the bar chart, it is clear that Ku Ring Gai Chase's trip count (1,277) far exceeds that of any other suburb. Although its resident population is small, trip volume is huge, mainly because:

1. The suburb covers a vast area, so the data aggregates a wide range of journeys under one name.
2. As a national park, Ku Ring Gai Chase attracts large numbers of overnight visitors each year.
3. Its remoteness and poor public transport make Uber the primary choice.

Similarly, other remote suburbs like Fiddletown also show higher fares—often exceeding \$40 per trip—due to long distances from the CBD and limited driver supply.

3.2 Affluent Coastal Tourism Hubs

Suburbs with average fares above \$40 also include Manly and Collaroy. As famous tourist destinations and affluent residential areas, their higher prices are driven by:

1. Large and stable tourist demand
2. Higher local willingness to pay—wealthy riders prefer comfortable, private travel
3. Limited driver supply

3.3 Fringe Suburbs with Moderate Fares – Glenorie, Kenthurst & Galston

These three suburbs lie on the city's edge and have average fares in the \$35–\$38 range. Trips here are primarily for commuting: moderate distances to the CBD, sufficient driver availability, and regular peak-hour patterns keep prices at a mid-level.

3.4 Peri-Urban Areas with Lower Fares – Rural, Ingleside & Belrose

These three areas record the lowest average fares among the top ten. This is mainly because they are close to major roads and public transport, making transit convenient and competitive. Local residents take short, everyday trips, and the presence of public-transport alternatives also pushes Uber prices down to attract riders.

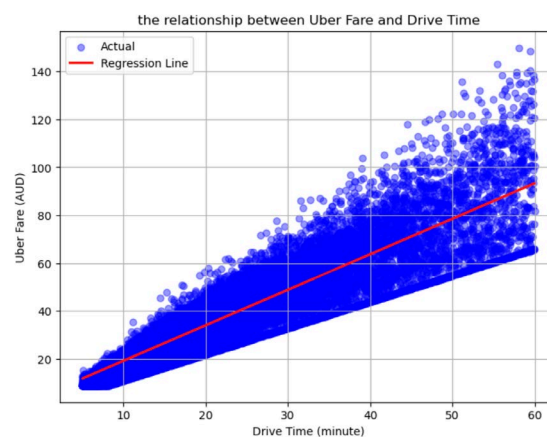
3.5 Conclusion

Geographic location has a significant impact on Uber fares. The farther from the city center, the poorer the road access and public transport, and the fewer the drivers—resulting in higher prices. By contrast, the city center benefits from short trips, strong competition, and abundant drivers, which keep fares relatively low and stable despite high demand. This study can help riders better understand how location affects fares and choose pickup points that offer the lowest prices.

4. Impacts of Drive Time and Drive Distance on Uber Fare

Given that travel time and driving distance are fundamental elements that cannot be ignored when passengers use Uber app, it is reasonable to speculate that these two factors may also play a key role in Uber's dynamic pricing mechanism. To further investigate the pricing dynamics of Uber rides, the following section examines how both drive time and drive distance influence fare levels with scatter plots and linear regression models.

4.1 Relationship between Drive Time and Uber Fare



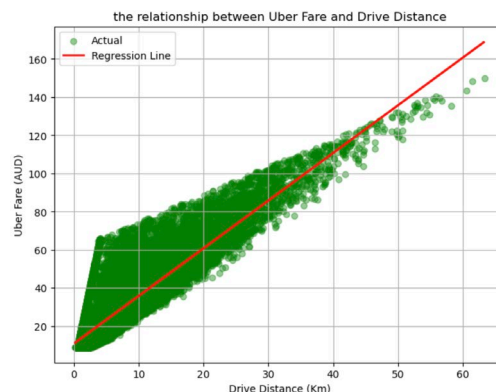
The above scatter plot was generated to relationship between the driving time and Uber price. It suggests a significant positive correlation between the Drive Time and Uber price. As the driving time increases, passengers need to charge more. Specifically, when the time is less than 30 minutes, the price variation among different trips is relatively small. And when driving time is greater than 30 minutes, the change of uber prices are greatly different. And there is a linear regression model to quantify this relationship. The regression equation is:

$$\text{Uber Fare} = 4.426 + 1.482 * \text{Drive Time}$$

For every 1-minute increase in Drive Time, Uber price will increase \$1.482. R square of this regression model is 0.836, which suggests that drive time is a significant predictor of Uber fare and plays a major role in the fare determination process.

This result reflects the time-based Uber pricing mechanism. Longer durations typically result in increased drivers' working time, prolonged vehicle usage and higher risks of traffic-related delays, all of which contribute to fare escalation. In particular, during periods of low speed or heavy traffic, the time-based pricing component can serve as a compensation mechanism for drivers. In such cases, the driving distance may be limited, but the time spent on the road is still high.

4.2 Relationship between Drive Distance and Uber Fare



As previous analysis, the distance of order is concentrated within the range of 5 to 15 km. Notably, when the driving distance is less than 20 km, significant variations of Uber fare can be observed. This indicates that shorter trips are subject to greater price fluctuations, which may be influenced by factors such as traffic conditions, route efficiency, and local surcharges. However, once the driving distance exceeds 20 km, the fare values tend to stabilize, suggesting the presence of a more consistent pricing

structure for longer trips.

Furthermore, a clear positive relationship is observed between driving distance and Uber fare, and the linear regression model is following:

$$\text{Uber Fare} = 10.908 + 2.497 * \text{Drive Distance}$$

Based on this regression model, for every 1km increase in driving distance, Uber price will increase \$2.497, which aligns with the basic principle of pricing models that account for both distance and time. This relationship suggests that the cost to the passenger is directly proportional to the amount of distance covered during the trip. Longer trips inherently involve higher fuel costs, increased vehicle wear and tear, and extended driver compensation time, all of which contribute to the higher fare.

4.3 The Combined Effect of Drive Time and Distance on Uber Fare

Because Uber users typically consider both the time and distance of their trips when planning travel and estimating costs, incorporating both Drive Time and Drive Distance as independent variables in a linear regression model offers a more comprehensive and realistic approach to fare prediction. The resulting regression equation is as follows:

$$\text{Uber Fare} = 0.078 + 0.999 * \text{Drive Time} + 1.447 * \text{Drive Distance}$$

The R square of this regression model is 0.999, which is relatively high due to the cleaning data, but still shows a strong fitting ability. Importantly, the regression coefficients reveal meaningful insights into the relative influence of each variable. The coefficient of 1.447 for Drive Distance is notably higher than the 0.999 for Drive Time, suggesting that distance has a greater impact on pricing than time in Uber's fare structure. This may reflect the fact that Uber's base pricing algorithm places greater weight on the physical length of the trip.

4.3 Explanations of Outliers

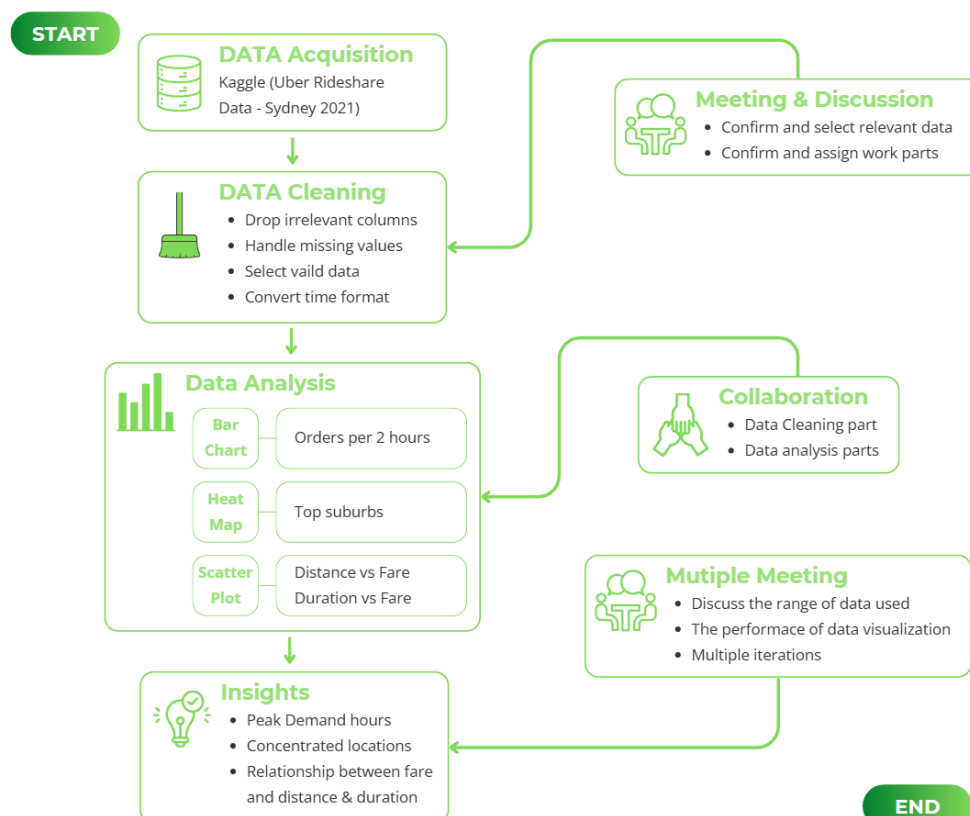
From the two scatter plots presented above, it is evident that certain passengers were charged fares significantly higher than the average. This deviation can be attributed to several key factors beyond drive time and distance.

The first factor is rush hour pricing. As established in earlier analyses, Uber employs a dynamic pricing algorithm (surge pricing), which adjusts fares based on real-time supply and demand conditions. During peak commuting hours, such as 7:00–9:00 a.m. in school zones, demand often exceeds supply, resulting in elevated fares. Conversely, during off-peak hours like 3:00–5:00 a.m. at midnight, fares tend to decrease due to lower demand and increased driver availability.

The third factor is pick-up area. The central areas typically have a higher density of available drivers compared to suburban areas. As a result, ride requests in central zones are more likely to be fulfilled promptly and at standard rates. In contrast, passengers in suburban or less densely populated areas may experience longer waiting time and increased fares due to limited driver availability.

Lastly, vehicle type is another potential determinant of fare variation. Different Uber service tiers (e.g., UberX, UberXL, Uber Comfort, or Uber Green) come with varying pricing structures. For instance, Uber Green, which utilizes electric or hybrid vehicles, generally offers lower fares due to reduced operating costs, while premium services like UberXL charge higher rates due to larger vehicle capacity and comfort features.

5. Flowchart:



6. Conclusions:

Based on the results of all the above bar charts, heat maps and scatter plots, we can find that Uber's pricing system can flexibly adjust its fees when facing changes in demand, region, time and distance, etc. Therefore, we draw the following three conclusions and analyses to support it.

6.1 Peak Demand Periods

By comparing the number of orders and order prices in different time periods, we found that the Uber fare during school zones and peak commuting hours was higher than that in other time periods. From this result, we can analyze that the Uber platform has adopted the strategy of raising order prices during peak periods.

6.2 Concentrated Locations

To further discuss the changes made by the Uber platform in response to the fluctuations in supply and demand levels, we have drawn a hot spot map of the starting points of Uber orders. Meanwhile, we have also respectively produced the top ten most popular areas in terms of order quantity and order price. By comparison, we were surprised to find that the order prices in remote areas with lower demand were relatively high, while those in central areas with huge demand were relatively ordinary. From this result, we can analyze that the Uber platform has invested a large amount of its resources in the central areas, while in remote areas, it uses high order prices to make up for the lack of quantity.

6.3 Distance and Duration

Finally, we will discuss the two most intuitive factors, time and distance. Whether observing the hot maps we completed before or from the perspective of life experience, the order time and distance are positively correlated with Uber's cost. And through regression analysis, we found that the Uber platform regards the order distance as the main role.

To sum up, through the analysis of data presentation and visual graphics, we have concluded that the dynamic pricing system of the Uber platform has the ability to flexibly change according to demand, location and order characteristics.

Reference

- [www.kaggle.com](https://www.kaggle.com/datasets/eugenekoutiloff/synthetic-dataset-of-uber-trips-in-sydney-nsw). (n.d.). Synthetic dataset of Uber trips in Sydney nsw. [online] Available at: <https://www.kaggle.com/datasets/eugenekoutiloff/synthetic-dataset-of-uber-trips-in-sydney-nsw>.
- ChatGPT (2025) Sydney Uber pick-up demand heat map [interactive map]. Generated using Uber ride data and Folium. OpenAI. Available at: <https://chat.openai.com> [Accessed 12 Apr. 2025].